ISF JUNICORN COHORT 3 • MAY 2026

Synthetic Biology

Solving the Data Center Cooling Crisis Through Synthetic Biology

Team iGEM IIT Bombay

IIT Bombay Research Park • Paris Grand Jamboree 2026

Synthetic Biology: Writing Life

Moving beyond just "reading" DNA (Genomics), Synthetic Biology allows us to **"write"** and program biological systems like computers.



Standardized

Using modular "BioBricks" to build biological circuits.



Micro-Factories

Programming cells to produce high-value chemicals.

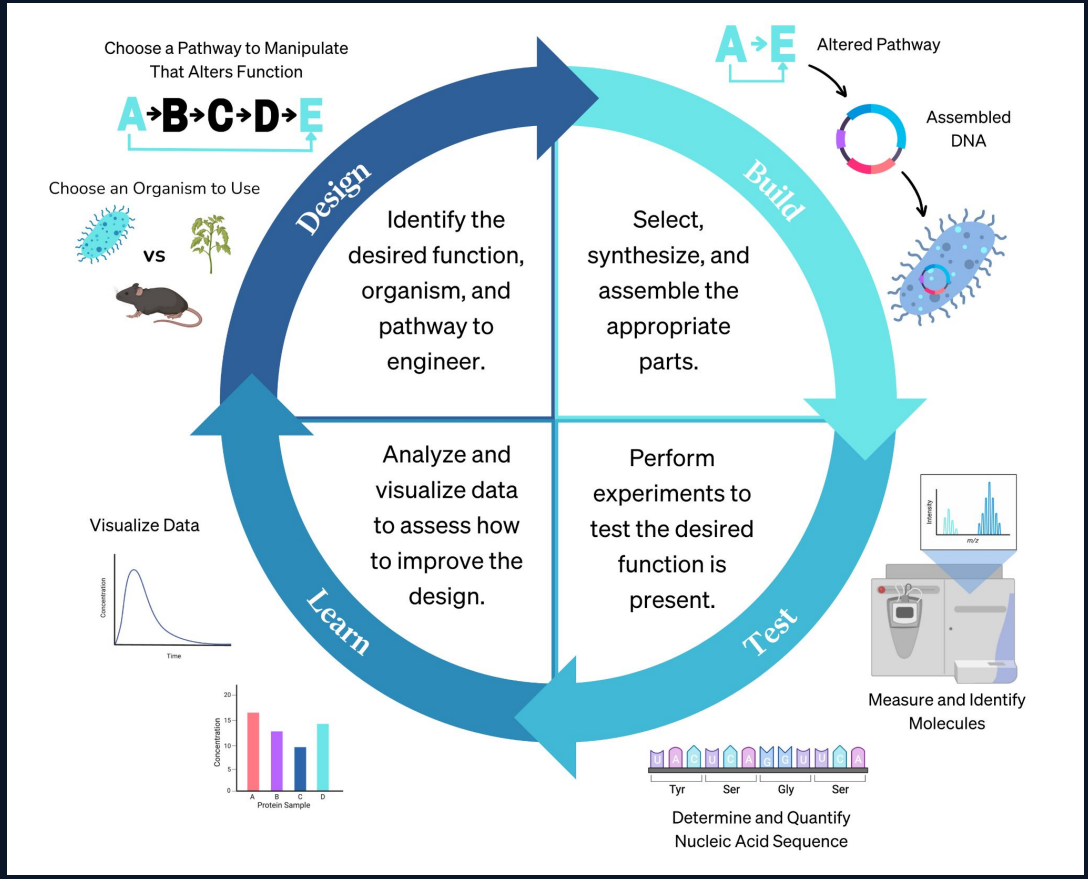


Sustainable

Replacing energy-intensive chemistry with ambient fermentation.

Synthetic Biology: Writing Life

Moving beyond just "reading" DNA (Genomics), Synthetic Biology allows us to **"write"** and program biological systems like computers.



SynBio: A Global Industrial Shift

1. Biocon – Bengaluru
2. Serum Institute of India – Pune
3. Panacea Biotec – New Delhi
4. Bharat Biotech – Hyderabad
5. Shantha Biotechnics – Hyderabad
6. Intas Biopharmaceuticals – Ahmedabad
7. Wockhardt – Aurangabad
8. Dr. Reddy's Laboratories (Biologics Division) – Hyderabad
9. Zydus Lifesciences (Biotech) – Ahmedabad
10. Reliance Life Sciences – Navi Mumbai
11. Lupin Biotech – Pune
12. Gennova Biopharmaceuticals – Pune
13. Laurus Bio – Bengaluru
14. Avesthagen – Bengaluru
15. Biological E. Limited – Hyderabad
16. Bharat Serums and Vaccines – Mumbai
17. Sanofi India (Shantha Unit) – Hyderabad
18. Syngene International – Bengaluru
19. Mitra Biotech – Bengaluru
20. Stempeutics Research – Bengaluru
21. Virchow Biotech – Hyderabad
22. Krebs Biochemicals & Industries – Hyderabad
23. Indian Immunologicals – Hyderabad
24. Cadila Healthcare Biotech Division – Ahmedabad
25. Hester Biosciences – Ahmedabad



The Engine: Team Subsystems



Wet Lab

Experimental execution: Plasmid design, cloning, and protein purification to turn theory into biological reality.



Dry Lab

Computational modeling: Molecular docking and data visualization to predict biological performance before testing.



Human Practices

Engaging stakeholders, policy experts, and the community to ensure our technology is responsible and safe.

Legacy: Project CalciCapture

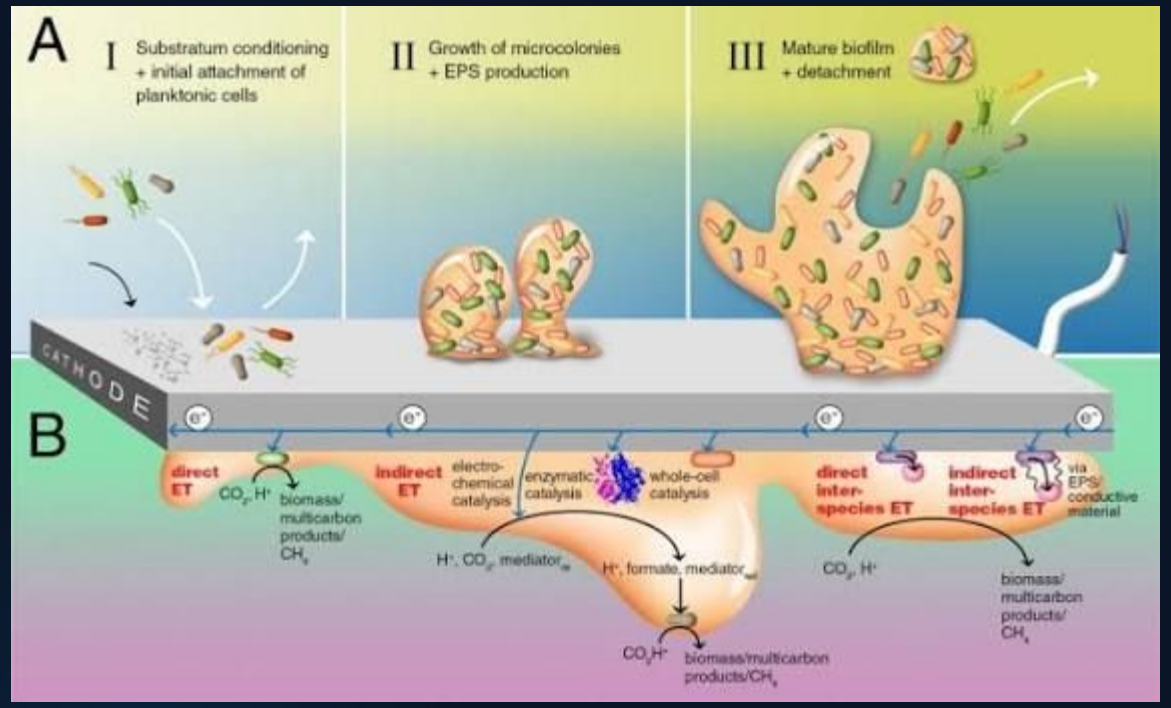
2024 iGEM Paris - Gold Medal

Solving the climate crisis through ocean-based carbon sequestration.

- Engineered *E. huxleyi* algae to increase carbon capture efficiency.
- Mimicked virus-induced calcification to trap CO2 in permanent mineral shells.
- Estimated 417 tonnes of carbon captured per plant annually.



Legacy: Project Aureolyze



2025 iGEM Paris - Silver Medal

Combatting Antibiotic Resistant Biofilms in critical healthcare settings.

- Engineered ESP protease to safely degrade tough bacterial biofilms.
- Used site-directed mutagenesis to ensure zero toxicity to human tissues.
- Won "Best Poster" at All India iGEM Meet (AIIM).

The Current Frontier

The Data Center Cooling Crisis

A Growing Thermal Crisis

30%

TOTAL ENERGY FOR COOLING

Reaching the Physical Wall

AI workloads and high-performance GPUs generate extreme heat. A single rack can hit 50kW, where traditional air cooling physically fails.

- Legacy fluids rely on toxic PFAS "forever chemicals."
- Existing bio-oils sludge and throttle server performance.

Methyl Myristate: The Bio-Coolant

Harnessing Liquid Intelligence

Our solution is a sustainable, high-performance dielectric ester produced via precision fermentation.

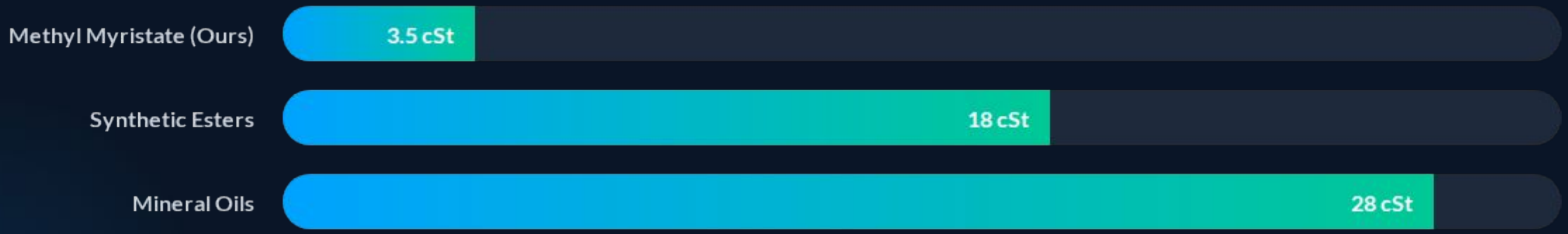
- 100% Biodegradable & PFAS-free.
- 5.4x higher thermal conductivity than air.
- Produced at 30°C (vs 450°C chemical synthesis).



Performance Comparison

Yarrowia lipolytica: The Cell Factory

We metabolically engineer oleaginous yeast to upcycle crude industrial glycerol into ultra-pure bio-coolant.



Lower viscosity (cSt) ensures lower pumping power & superior heat dissipation.

Sponsorship Tiers: Paris 2026

Tier	Investment	Key Benefits
Platinum	₹3,00,000	Lab naming rights, Paris booth branding, Chief Guest keynote slot.
Gold	₹2,00,000	Major jersey branding, monthly newsletter spotlight, joint CSR reports.
Silver	₹1,00,000	Team video recognition, lab coat logo, social media feature day.
Bronze	₹50,000	Technical citation in research archive, digital credits on website.

Questions?

Join us at the frontier where engineering meets biology.

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